

Motif Statistics in Optimized Inventory Policies

DeepSeek + SciPy EoH Runs

May 2026

Abstract

This report summarizes structural motifs in optimized inventory-control policies. The analysis uses a conservative static detector applied to saved population files, focusing on high-coverage DeepSeek+SciPy runs. The main finding is that policies retain an order-up-to/base-stock backbone while increasingly adding partial adjustment, pipeline weighting, and order clipping across generations. Distribution families show different dominant refinements: exponential instances emphasize smoothing and pipeline timing, Poisson instances emphasize clipping, and normal instances retain the strongest order-up-to structure.

1 Scope and Data

The analysis uses policies from `pops/population_generation_*.json`, not prompt files, because these files provide explicit generation identifiers. The scope is:

- Model and optimizer: DeepSeek + SciPy, operator `m2`.
- Generations: 1 through 10.
- Distributions: the 14 distributions whose prompt-cost matching rate exceeded 80% under the previous 1% same-distribution matching rule.
- Sample size: 5,052 policy records from 53 run folders.

2 Conservative Motif Detector

The detector uses Python AST parsing and dependency tracking from the final order expression. It is intentionally conservative:

- **Inventory position** requires an unweighted combined signal using on-hand inventory and total pipeline inventory.
- **Order-up-to** requires a positive-part target-minus-state gap, such as `max(0, target - state)`.
- **Partial adjustment** requires a gain or smoothing parameter multiplying a gap/order response.
- **Pipeline weighting** requires weighted pipeline terms on the dependency path to the final order decision.
- **Constant order** requires an additive baseline in the final order expression, not merely safety stock inside a target level.

- **Order clipping** detects upper/lower bounding beyond the positive-part operator.
- **Thresholding** requires a meaningful state/order threshold that controls the order formula, excluding implementation checks such as `len(pipeline_orders)>0`.

3 Overall Motif Frequencies

Average number of motifs per policy: **3.43**.

Table 1: Overall motif frequencies

Motif	Frequency
Inventory position	81.6%
Order-up-to	87.1%
Partial adjustment	55.1%
Pipeline weighting	43.5%
Constant order	14.4%
Order clipping	52.7%
Thresholding	9.0%

Interpretation. The order-up-to motif remains the dominant backbone. However, more than half of the policies include partial adjustment or clipping, and roughly 43% use some form of pipeline weighting. This indicates that optimization does not simply replace base-stock logic; it layers control refinements on top of a target-seeking structure.

4 Generation-Wise Pattern

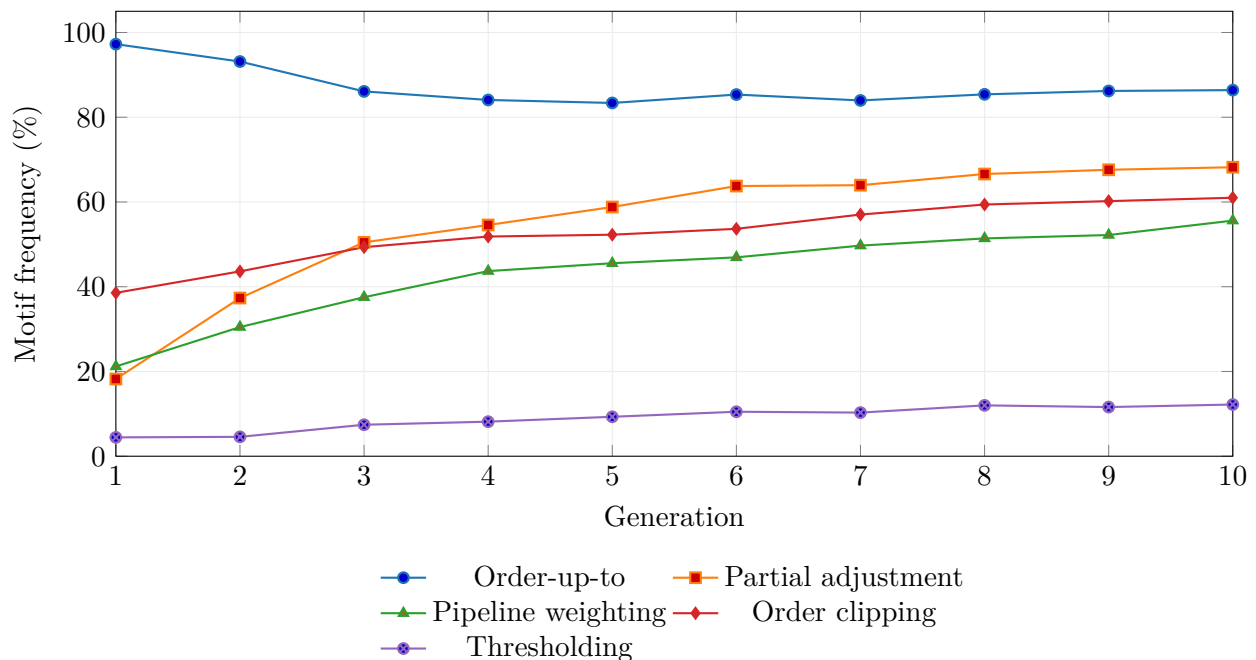


Figure 1: Motif frequencies by generation.

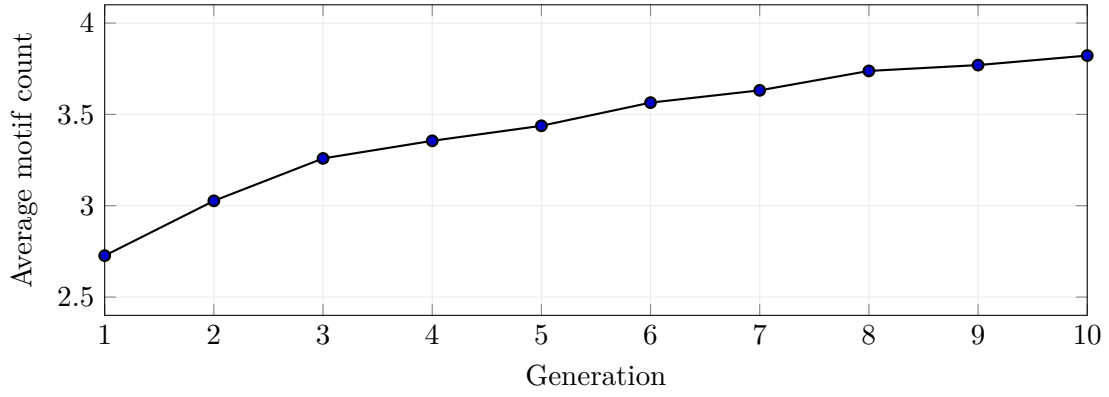


Figure 2: Average number of detected motifs per policy.

Table 2: Selected generation-wise statistics

Gen	N	Avg. motifs	Order-up-to	Partial	Pipeline	Clipping	Threshold
1	472	2.73	97.2	18.2	21.2	38.6	4.4
2	525	3.03	93.1	37.3	30.5	43.6	4.6
3	525	3.26	86.1	50.5	37.5	49.3	7.4
4	515	3.36	84.1	54.6	43.7	51.8	8.2
5	505	3.44	83.4	58.8	45.5	52.3	9.3
6	505	3.56	85.3	63.8	46.9	53.7	10.5
7	505	3.63	84.0	64.0	49.7	57.0	10.3
8	500	3.74	85.4	66.6	51.4	59.4	12.0
9	500	3.77	86.2	67.6	52.2	60.2	11.6
10	500	3.82	86.4	68.2	55.6	61.0	12.2

From generation 1 to generation 10, average motif count increases from 2.73 to 3.82. The largest changes are:

- Partial adjustment: +50.0 percentage points.
- Pipeline weighting: +34.4 percentage points.
- Order clipping: +22.4 percentage points.
- Thresholding: +7.8 percentage points.

Table 3: Motif-count distribution in generation 1 vs. generation 10

Number of motifs	Gen. 1 share (%)	Gen. 10 share (%)
0	0.0	0.0
1	1.3	1.2
2	44.5	10.8
3	38.8	28.6
4	11.4	29.2
5	3.8	24.4
6	0.2	5.8
7	0.0	0.0

Takeaway. Early policies are mostly simple 2–3 motif rules. By generation 10, mass shifts toward 4–6 motif policies, consistent with progressive structural enrichment.

5 Distribution-Wise Differences

Table 4: Family-level weighted averages

Family	N	Avg. motifs	Order-up-to	Partial	Pipeline	Clipping	Threshold
Exponential	1431	3.63	87.9	64.1	56.0	45.2	7.8
Poisson	3124	3.34	84.7	52.6	37.7	54.8	8.8
Normal	497	3.46	99.4	44.7	43.7	61.4	14.3

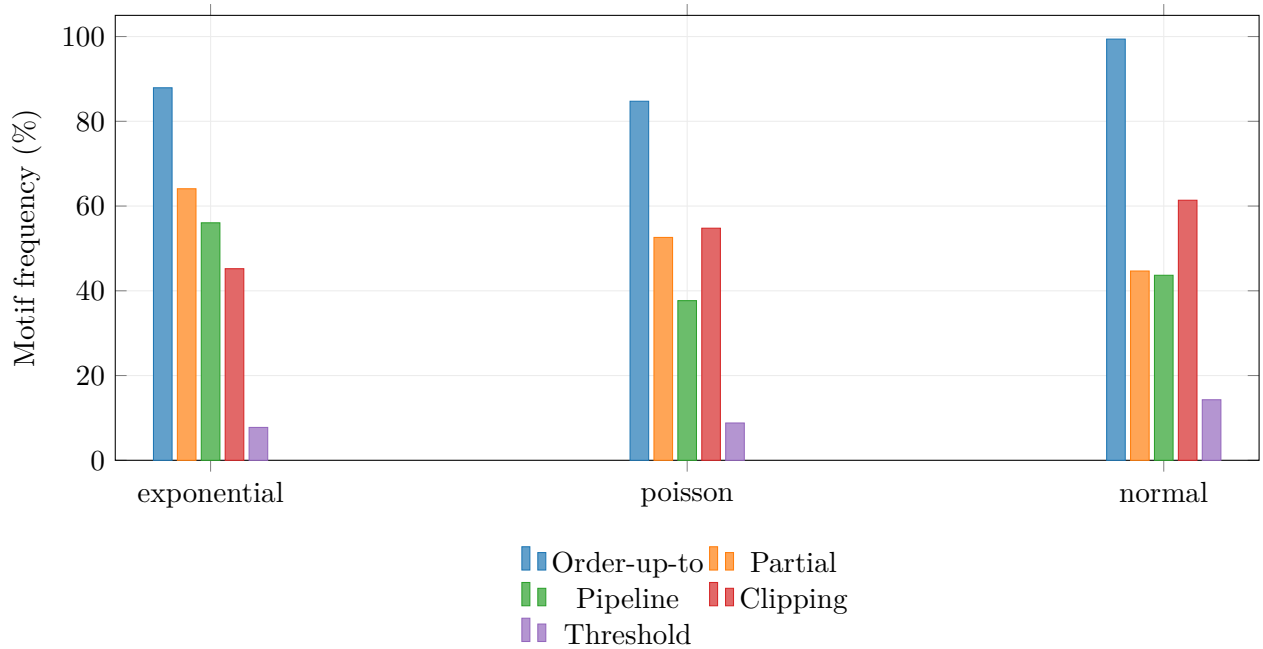


Figure 3: Motif frequencies by distribution family.

Table 5: Distribution-level motif frequencies

Distribution	N	Avg.	Inv. pos.	Order-up	Partial	Pipeline	Clipping	Threshold
exponential_L2_c1_2	200	2.98	97.0	61.0	35.0	46.5	39.5	12.5
exponential_L2_c1_5	195	3.65	100.0	76.4	65.6	59.0	50.3	11.8
exponential_L4_c1_2	200	3.75	85.0	91.5	83.0	54.5	51.5	4.5
exponential_L4_c1_5	298	3.98	100.0	100.0	81.9	39.9	50.7	4.0
exponential_L6_c1_2	298	3.21	65.8	92.6	39.3	61.1	33.6	7.7
exponential_L6_c1_5	240	4.13	87.1	95.8	80.0	76.7	48.3	7.9
normal_std30_L6_c1_2	100	2.86	12.0	98.0	0.0	32.0	88.0	48.0
normal_std30_L6_c1_5	397	3.61	95.7	99.7	55.9	46.6	54.7	5.8
poisson_L2_c1_2	298	3.34	94.6	93.6	81.2	43.6	14.8	5.4
poisson_L2_c1_5	297	3.60	66.7	96.3	70.7	71.7	45.5	8.8
poisson_L4_c1_2	129	3.71	60.5	96.1	48.1	41.9	65.1	40.3
poisson_L4_c1_5	494	3.40	88.9	66.0	70.9	26.3	67.8	3.6

Distribution	N	Avg.	Inv. pos.	Order-up	Partial	Pipeline	Clipping	Threshold
poisson_L6_c1_2	1111	3.03	72.2	82.9	29.4	42.0	44.8	8.9
poisson_L6_c1_5	795	3.59	84.3	89.4	56.9	23.0	77.4	8.1

6 Main Messages for Presentation

1. **The base-stock backbone remains visible.** Order-up-to appears in 87.1% of policies overall, and inventory-position structure appears in 81.6%.
2. **Optimization enriches rather than replaces the backbone.** Later generations increasingly add partial adjustment, pipeline weighting, and clipping.
3. **Structural complexity increases by generation.** Average motif count rises from 2.73 in generation 1 to 3.82 in generation 10.
4. **Different demand families favor different refinements.** Exponential instances have stronger partial adjustment and pipeline weighting; Poisson instances have stronger clipping; normal instances retain the strongest order-up-to structure and relatively more thresholding.
5. **The detector is conservative.** Some complex policies may be under-counted, but the reported trends are less likely to be driven by false positives from implementation details.

7 Limitations

The motif labels are produced by static code analysis. They should be interpreted as structural evidence, not formal semantic equivalence. The detector intentionally avoids ambiguous cases such as implementation checks and target-level safety stock when labeling thresholding or constant-order motifs. A small manual audit file is included with positive and negative examples for each motif.